Lecture Notes on

QFT

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Resources used:

- Xi Yin's 253ab courses.
- Steven Weinberg's two volumes.
- Daniel Harlow's notes.
- Sidney Coleman's QFT.
- Peskin and Schroeder.
- Schwartz.
- Srednicki.
- *FIELDS* by Warren Siegel.

We mostly follow Weinberg's notation, except where his notation sucks.

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1 Overview

(overview things here; talk about how single-particle rel qm is inconsistent; talk about how particles are fundamental, and not fields (not so sure about this anymore))

Which is more fundamental, particles or fields? Most physicists are likely to say fields—they are *wrong*: particles are fundamental. To see this, just ask an experimentalist what they measure. They measure *particles*! In the calorimeter at the LHC, *particles* hit the walls of the collider, blazing paths through the detector that we can reconstruct as paths these particles take. The energy and momentum of these particles, combined with the ingenuity of experimentalists, tells you exactly what kind of particle this was. There is no mention of fields anywhere in this picture. **(can steelman this, but should also fight back)**

(think about this more)

An "axiom" of QFT is that it has local, microcausal field operators. These represent local "measurement" on our system. One can think of this as taking inspiration from electromagnetism, since that's our only example of a truly local observable (e.g. the electric field; you measure it at a *point*. This is obvious actually, as this is what *must* occur mathematically, i.e. $\phi\colon M \to \mathbb{R}$, so there is no sense of asking its value at anywhere but a point you "can reach"). Since these are, by definition, observables, they also must be Hermitian. Since things are local, they must obey Poincaré symmetry. Enumerating these conditions for the scalar field gives

1. **Poincaré symmetry**: under a Poincaré transformation $U(\Lambda, a)$, our fields must transform as
$$U(\Lambda, a)\phi(x)U(\Lambda, a)^{-1} = \phi(\Lambda x + a).$$
2. **Microcausality**: our fields must be local and causal, e.g. spacelike separated measurements can't effect each other. This corresponds to
$$[\phi(x), \phi(y)] = 0 \qquad\qquad (x-y)^2 > 0.$$
3. **Unitarity**: as stated, the fields must be Hermitian, $\phi = \phi^\dagger$.

2 Relativistic Quantum Mechanics

We start by recalling the axioms of quantum mechanics[1]:

Idea 2.1 (Axioms of quantum mechanics)

The axioms of quantum mechanics are:

- Quantum mechanical **states** are represented by **rays** in a **Hilbert space**. We will denote Hilbert spaces by $\mathcal{H}$.
- **Observables** are represented by self-adjoint operators on our Hilbert space. Observed quantities are the eigenvalues of vectors in $\mathcal{H}$.

[1] Of course, there are different axioms than this that give identical results. "Axioms" is really a misnomer; they're just assumptions that let us build the rest of the theory. There is nothing intrinsically fundamental about this set of axioms as opposed to other ones.

- Let $\mathcal{R}$ be a ray. If $\mathcal{R}_1, \ldots, \mathcal{R}_n$ are a set of mutually orthogonal rays, the probabilty of observing $\mathcal{R}$ in $\mathcal{R}_i$ is

$$\mathbb{P}(\mathcal{R} \to \mathcal{R}_i) = |\langle \psi | \phi_i \rangle|^2,$$

where $\psi \in \mathcal{R}$ and $\phi_i \in \mathcal{R}_i$.

Remarks:

- Recall that rays are equivalence classes of kets up to phase and normalization. Since this is an equivalence class, WLOG we can choose our representative to be normalized.
- Self-adjoint or Hermitian? **(comment on this)**
- In our third axiom, we make reference to experiment. How do we physically represent an experiment on our Hilbert space? **(good question)**

The most important kinds of operators are operators that realize *symmetry transformations* on our Hilbert space.

(finish)

2.1 Symmetries

(add the normal ones, and then ps in the next section)

(for whatever reason) , this continuity may also be found in the *representation* of the symmetry.

2.2 Poincaré symmetry and the little group

Since we are doing relativistic quantum mechanics, let's see how the relativistic symmetry group acts on our Hilbert space. The Poincaré group in mostly-plus is given by

$$\mathcal{P} = \mathbb{R}^D \rtimes SO(D-1, 1).$$

Consider some Poincaré transformation $(\Lambda^\mu{}_\nu, a^\mu)$. By Wigner's theorem, this is represented as a linear unitary operator on our Hilbert space; the group property is

$$U(\Lambda, a)U(\bar{\Lambda}, \bar{a}) = U(\Lambda\bar{\Lambda}, \Lambda\bar{a} + a).$$

Notice that this form implies that the Poincaré group is a *semi-direct product*, not a direct product (as the translation entry is "twisted" by Λ). Since the Poincaré group is a Lie group, things are continuous. Taylor-expanding $U(\Lambda, a)$ gives

$$U(\Lambda, a) \simeq \delta^\mu_\nu - ia_\mu \mathcal{P}^\mu + \frac{i}{2}\omega_{\mu\nu} J^{\mu\nu}.$$

The generators of $U(\Lambda, a)$ are very special: they are the four-momentum operator and the angular momentum/boost operator. Considering their components gives

$$P^\mu = (H, \mathbf{P}), \qquad J^{\mu\nu} = \begin{cases} J^{0i} = K^i \\ J^{ij} = \epsilon^{ijk} J_k, \end{cases}$$

where K^i is the generator of boosts.

We would now like to see how $U(\Lambda, a)$ acts on our states. Choose our states to be labeled as $|\mathbf{k}, \sigma\rangle$, where $\mathbf{k}$ is our spatial momentum and σ is some set of internal degrees of freedom (which have not yet been specified). Note that we may well use $|k, \sigma\rangle$ instead of $|\mathbf{k}, \sigma\rangle$, but we choose $|\mathbf{k}, \sigma\rangle$ here because it emphasizes that these particles are all on-shell (i.e., that $k^0 = \sqrt{|\mathbf{k}|^2 + m^2}$). First, note that this basis diagonalizes translations,

$$U(0, a)|\mathbf{k}, \sigma\rangle = e^{-i\mathcal{P}a}|\mathbf{k}, \sigma\rangle = e^{-ika}|\mathbf{k}, \sigma\rangle.$$

Indeed, this is one of the reasons we chose our states to be labeled by momentum eigenvalues; momentum is symmetric under translations. Let us now study how Lorentz transformations act on these states. *Define* an arbitrary state as

$$|\mathbf{k}, \sigma\rangle \coloneqq \mathcal{N}(\mathbf{k})U(L(\mathbf{k}))|\mathbf{k}_R, \sigma\rangle,$$

where $\mathbf{k}_R$ is a given **reference momentum**[2], $L(\mathbf{k})$ is some Lorentz transformation that takes $\mathbf{k}_R \xrightarrow{L(\mathbf{k})} \mathbf{k}$, and $\mathcal{N}(\mathbf{k})$ a normalization factor to be determined. Consider $U(\Lambda)|\mathbf{k}, \sigma\rangle$ for arbitrary Λ. We have

$$\begin{aligned} U(\Lambda)|\mathbf{k}, \sigma\rangle &= \mathcal{N}(\mathbf{k})U(\Lambda)U(L(\mathbf{k}))|\mathbf{k}_R, \sigma\rangle \\ &= \mathcal{N}(\mathbf{k})\mathcal{N}(\boldsymbol{\Lambda k})^{-1}U(L(\boldsymbol{\Lambda k}))U(\underbrace{L(\boldsymbol{\Lambda k})^{-1}\Lambda L(\mathbf{k}))}_{W})|\mathbf{k}_R, \sigma\rangle. \end{aligned}$$

Let's study W. The following diagram commutes:

$$\begin{tikzcd} \mathbf{k}_R \arrow[r, "L(\mathbf{k})"] \arrow[drr, "W"'] & \mathbf{k} \arrow[r, "\Lambda"] & \boldsymbol{\Lambda}\mathbf{k} \arrow[d, "L(\boldsymbol{\Lambda}\mathbf{k})^{-1}"] \\ & & \mathbf{k}_R \end{tikzcd}$$

Thus, W fixes $\mathbf{k}_R$. We call W a **Wigner rotation**, and we call the set of such W's the **little group** for our given $\mathbf{k}_R$. Note that the little group is specific to the kind of reference momentum $\mathbf{k}_R$ is, e.g. massless particles in $D = 4$ form one little group.

The key observation to make is that while $U(W)$ can't change $\mathbf{k}_R$, it can change our σ labels; thus, the σ labels tell you how the particle transforms under the *little group*. We may generically write such a representation of the little group as

$$U(W)|\mathbf{k}_R, \sigma\rangle = \sum_{\sigma'} D_{\sigma\sigma'}(W)|\mathbf{k}_R, \sigma'\rangle.$$

Notice that $D(W_1W_2) = D(W_2)D(W_1)$, as we can see from the definition. We now determine the normalization from orthogonality. Consider **(finish)** Putting this all together, we have that

$$\boxed{U(\Lambda)|\mathbf{k}, \sigma\rangle = \sqrt{\frac{\omega_{\boldsymbol{\Lambda}\mathbf{k}}}{\omega_{\mathbf{k}}}} \sum_{\sigma\sigma'} D_{\sigma\sigma'}(W(\mathbf{k}))|\boldsymbol{\Lambda}\mathbf{k}, \sigma'\rangle.}$$

[2]This will be specific to the kinds of particles you are talking about, e.g. $k_R = (E, 0, 0, E)$ is a reference momentum for the massless little group in $D = 4$.

(why does this hold for multiparticle states? I think because they are diff vectors, so doesn't matter; can use raising and lowering ops i think)

$$\boxed{U(\Lambda, a)|\mathbf{k}_1, \sigma_1; \mathbf{k}_2, \sigma_2; \ldots; \mathbf{k}_n, \sigma_n\rangle = \prod_{i=1}^{n} e^{-ik_i a} \sqrt{\frac{\omega_{\mathbf{\Lambda k}_i}}{\omega_{\mathbf{k}_i}}} \sum_{\sigma_i \sigma_i'} D_{\sigma_i \sigma_i'}(W(\mathbf{k}_i))|\mathbf{\Lambda k}_1, \sigma_1'; \mathbf{\Lambda k}_2, \sigma_2'; \ldots, \mathbf{\Lambda k}_n, \sigma_n'\rangle}$$

This will be useful when we talk about scattering later.

We now take a closer look at the D matrices. **(finish)**

2.3 Lagrangian QM

Now we'll get into Lagrangian QM, which naturally introduces the path integral and accompanying ideas (regularization, renormalization, etc.).

Idea 2.2 (The path integral)

Let $\langle q_f|$ and $|q_i\rangle$ respectively be final and initial position eigenstates. Call the time-translation operator for a time T, $U(T)$. **(finish)**

(add lagrangian qm; reg, renorm, path integral derivation; add exercises from yin and weinberg here too)

3 Classical Field Theory

We briefly review classical field theory here; for a more in-depth analysis, see the **(notes)** on classical field theory.

As always, we start with an action:

$$S[\phi] = \int d^D x\, \mathcal{L} = \int dt \int d^{D-1}\mathbf{x}\, \mathcal{L}(\phi, \partial\phi, \ldots),$$

where the bounds on this integral are generically over all of $\mathbb{R}^D$, but may be restricted to a subset of it (e.g. the upper half plane of $\mathbb{R}^2$). The equations of motion are derived from the action by demanding stationarity

$$\frac{\delta S}{\delta \phi} = 0 \iff \boxed{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0.}$$

If we interchange ϕ for a tensor field, e.g. A_μ or $h_{\mu\nu}$, the ELE above become tensor equations: one for each component of the tensor field.

(talk about canonical coordinates here)

We now investigate symmetries of the action—we consider the infinitesimal case and build from there. Let

$$\phi \to \phi'(x) = \phi(x) + (\delta_\epsilon \phi)(x),$$

where $(\delta_\epsilon \phi)(x) := \epsilon\, \delta\phi(x)$, with $\epsilon \ll 1$ and $\delta\phi(x)$ a field variation. Notice only the field variable is primed. If we prime both variables, we get an equality: $\phi'(x') = \phi(x)$. This is fundamentally because

the transformed field at the transformed point should give the same measurement as the original field at the original point. If the coordinate transformation above is a symmetry, that means that the action is unchanged under such a transformation. Consider a non-symmetry transformation:

$$\phi \to \phi'(x) = \phi(x) + \tilde{\delta}_\epsilon \phi(x),$$

where $(\tilde{\delta}_\epsilon \phi)(x) := \epsilon(x)\delta\phi(x)$. We call this process of making ϵ x-dependent "gauging the symmetry"—we also say this is turning a global symmetry into a local one. The action should no longer be invariant under such a transformation, but should take the form

$$\tilde{\delta} S = \int d^D x\, \partial_\mu \epsilon(x) j^\mu(x) \xrightarrow{\epsilon(x)=\epsilon} \delta S = 0.$$

Note that this implies that $\partial_\mu j^\mu = 0$ when ϵ is constant, thus giving us a locally conserved current. Integrating $\tilde{\delta} S$ by parts gives an expression for $j^\mu(x)$,

$$\boxed{j^\mu(x) = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi - \mathcal{U},}$$

where $\mathcal{U}$ is what's left over. Concretely,

$$\begin{aligned}
\tilde{\delta} S[\phi] &= \int d^D x\, \frac{\partial \mathcal{L}}{\partial \phi} (\tilde{\delta}_\epsilon \phi)(x) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu (\tilde{\delta}_\epsilon \phi)(x) \\
&= \int d^D x\, \partial_\mu \epsilon(x) \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi(x) \right) + \epsilon(x) \underbrace{\left(\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi(x) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu \delta\phi(x) \right)}_{\partial_\mu \mathcal{U}(x)}.
\end{aligned}$$

We may also form a conserved charge from j^μ by integrating its time component over some fixed time-slice. We have

$$Q := \int d^{D-1}\mathbf{x}\, j^0(\mathbf{x}).$$

One can see that Q is conserved by taking its time derivative and using the fact that $\partial_\mu j^\mu = 0$. One can also check that Q generates the symmetry $\delta\phi$ in the sense of Hamiltonian mechanics:

$$\delta\phi(x) = \{Q, \phi\}|_{\pi(\mathbf{x}) = \partial\mathcal{L}/\partial\dot{\phi}(\mathbf{x})}$$

The most important Noether current in a classical field theory is the stress-energy-momentum tensor $j^\mu(x) = a^\nu T^\mu{}_\nu(x)$. Consider its derivation: if we translate the fields by an infinitesimal amount a^μ, we have that

$$\begin{aligned}
\phi'(x') &= \phi(x) \\
\phi'(x+a) &= \phi(x) \\
\phi'(x) + a^\mu \partial_\mu \phi'(x) &= \phi(x) \\
\implies \phi'(x) &= \phi(x) - a^\mu \partial_\mu \phi(x).
\end{aligned}$$

So, $\epsilon \rightsquigarrow a^\mu$ and $\delta\phi \rightsquigarrow \partial_\mu \phi(x)$. Note an important step: we replaced $\partial_\mu \phi'(x)$ with $\partial_\mu \phi(x)$. We can do this because $\phi'(x) = \phi(x)$ to first-order: concretely,

$$a^\mu \partial_\mu \phi'(x) - a^\mu \partial_\mu \phi(x) = a^\mu \partial_\mu (\phi'(x) - \phi(x)) \simeq a^\mu \partial_\mu (\delta_a \phi(x)) \sim O(a^2),$$

and thus we're safe to replace $\partial_\mu \phi'(x)$ with $\partial_\mu \phi(x)$. Inputting these variations into the expression for $j^\mu(x)$ gives

chung

Example 3.1 (Poincaré symmetry)
(todo)

4 Spin-0 QFTs

5 Scattering

We now get into scattering theory. This is probably the most important part of QFT, so we'll go relatively deep into it.

5.1 The basics

Composite particles decay, and fundamental particles **(finish)**

5.2 Symmetries of the S-matrix

5.3 The LSZ reduction

Remark 5.1 (On our derivation)**.** For the LSZ reduction, we follow Haag and Ruelle by way of Yin. You may see his lecture notes on this from 253a linked on the first page of these notes.

We will purposefully state many approximate results. One can get true equalities here, but the derivation is much more obtuse, so we will stick to our wave packets. Also, everything is basically the same anyways, so I don't care.

The goal of the LSZ reduction is to write our in and out states in terms of our field operators, thus giving us a connection between the S-matrix and perturbation theory. A priori, there is no connection here to speak about—indeed, the LSZ reduction is **the most important result in all of quantum field theory**.

We begin by defining a "smeared field operator" via

$$\phi_f \coloneqq \int d^D x\, f(x)\phi(x).$$

We assume our smearing function's Fourier transform is supported near the mass shell:

$$\tilde{f}(k) \neq 0 \xleftrightarrow{\approx} k^2 + m^2 = 0.$$

We must be able to talk about "moving fields"—how else are they supposed to travel out to infinity? Consider the following transformation on f:

$$f(k) \mapsto f^{(T)}(k) \text{ by } f^{(T)}(k) = e^{i(k^0-\omega_{\mathbf{k}})T} f(k).$$

How does this transformation act on our field operators? **(add wave equation part here)** Furthermore, the support of $\tilde{f}$ kills the multi-particle contribution of $\phi_f|\Omega\rangle$. Recall

$$\phi(x)|\Omega\rangle = \int \bar{d}^{D-1}\mathbf{k}\, e^{-ikx}\, \mathcal{Z}^{\text{eff}}_{\mathbf{k}}|\mathbf{k}\rangle + \int_{\text{m.p.}} d\alpha\, e^{-ip_\alpha x}\, \mathcal{Z}^{\text{eff}}_{\alpha}|\alpha\rangle$$

Applying the definition of ϕ_f and taking the inverse Fourier transform of f gives

$$\begin{aligned}\phi_f|\Omega\rangle &= \int d^Dx \int đ^{D-1}\mathbf{k}\, e^{-ikx}\, \mathcal{Z}_{\mathbf{k}}^{\text{eff}}|\mathbf{k}\rangle + \int d^Dx \int_{\text{m.p.}} d\alpha\, e^{-ip_\alpha x}\, \mathcal{Z}_\alpha^{\text{eff}}|\alpha\rangle \\ &\approx \int đ^{D-1}\mathbf{k}\, \tilde{f}(\mathbf{k},\omega_{\mathbf{k}})\, \mathcal{Z}_{\mathbf{k}}^{\text{eff}}|\mathbf{k}\rangle + 0\end{aligned}$$

(explain why T transformation doesn't do anything) Thus,

$$\boxed{\phi_f|\Omega\rangle \approx \phi_{f^{(T)}}|\Omega\rangle.}$$

The intuition behind this is that "there are a lot more operators than states". Thus we may define our states as

$$|\phi_{f_1},\ldots,\phi_{f_n}\rangle \coloneqq \phi_{f_n}\cdots\phi_{f_1}|\Omega\rangle.$$

Thus, by taking inner products of our states, we may get the S-matrix in terms of field operators acting on the vaccuum. We have **(add T variables on all of these guys below)**

5.4 Analyticity properties of the S-matrix

(my goat xi yin)
(unitarity bounds too?)

6 Spin-1/2 QFTs

7 Spin-1 QFTs

(motivation)

7.1 Gauge reundancy; $2 \neq 3$

There are two cases for spin-1 particles: massless or massive. They correspond to different representations of the little group, respectively being $SO(D-1)$ and $SO(D-2)$. In $D=4$, this means that a massive spin-1 particle has *three* spin DOF, while a massless spin-1 particle has *two* spin DOF. This discontinuous jump in the number of DOF between massive and massless is the root cause behind the difficulties of dealing with massless spin-1 particles.

Concretely, we describe the potential of (say) a photon using a four-vector to manifest Poincaré symmetry, i.e.

$$A_\mu(x) = \epsilon_\mu e^{ikx}.$$

We can demand transversality, $k\cdot\epsilon = 0$, which takes us from $4 \to 3$ DOF. If we have a massive particle, hooray! $3 = 3$. But if we have a massless particle, we're dead in the water—there are no other Lorentz invariant constraints that we can impose on ϵ. Thus, we have introduced *gauge redundancy*: concretely, we have defined the equivalence relation

$$\epsilon_\mu \sim \epsilon_\mu + \alpha k_\mu.$$

Thus, saying "the" polarization vector when referring to a photon is a misnomer—you really mean "a" polarization vector. We can see from this example that the biggest qualitative difference between massless and massive spin-1 QFTs is the number of DOF of their particles. This will cause lots of problems for us later, so buckle in and let's go!

(general Lorentz transformation properties of spin-1 particles go here;)

(

- **particles; massive vs massless**
- **little group representations**
- **particles -¿ fields; massless vs massive**
- **Path integral construction; gauge fixing**

)

8 Spin-3/2 QFTs

(susy...)

9 Spin-2 QFTs

(gravity!)

(maybe I should find a better organization scheme for this)

10 Useful formulae

10.1 Integrals and special functions

(schwinger parameters, feynman parameters, equivalence of the two)

Schwinger Parameters

We have that

$$\int_0^\infty ds\, s^{-a-1} e^{-bs-c/s} = 2\left(\frac{b}{c}\right)^{1/2} K_a(2\sqrt{bc}).$$

10.2 Dirac matrices

We reproduce some of the results covered in the notes on **(group theory)** here. We may form the Dirac matrices in any dimension. The most useful is $D = 4$. A choice of Dirac matrices in $D = 3+1$ is

$$\gamma^0 = -i\begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix} \qquad \gamma^a = -i\begin{pmatrix} 0 & \sigma^a \\ -\sigma^a & 0 \end{pmatrix}.$$

You can construct the Dirac matrices in any dimension by simply finding matrices that obey the Clifford algebra, $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$. Practically, this is done by bootstrapping your way up from a given dimension via the following iterative process:

- **Even → odd**: to go from even to odd dimensions, add the chirality matrix to your basis, $\gamma_c = \alpha\gamma^1 \cdots \gamma^n$. Fix α by demanding $(\gamma_c)^2 = 1$. This gives two inequivalent irreps, $\{\gamma^{(n)}, \gamma_c\}$ and $\{\gamma^{(n)}, -\gamma_c\}$.
- **Odd → even**: to go from odd to even dimensions, tensor the existing representation by one Pauli matrix, and add the identity times another Pauli matrix to get your representation. Concretely, if your odd rep is $\gamma^{(2k+1)}$, then your even rep is $\{\gamma^{(2k+1)} \otimes \sigma^a, I_{2k} \otimes \sigma^b\}$.